WELCOME TO
ISLAMIC UNIVERSITY OF TECHNOLOGY

Hum 4147
Technology, Environment and Society

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www.iutoic-dhaka.edu
Creating Global Citizens Rooted with Islamic Values

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Technology, Environment & Society

Course Learning Outcomes (COs)

After completing the course student will be able to

CO1: Explain the interdependency among technology, environment, society, and sustainable development;

CO2: Explain the current state and trends of technological growth, its contribution to human development and its use as an instrument of change in the 21st century;

CO3: Assess the impact of technology on environment and the impact of the environment upon human changes in the global climates;

CO4: Identify the environment friendly technologies, inc. renewable energy and environments and explain how they are environment friendly.

CO5: Identify the major hazards of industry and illustrate the ways to improve the working conditions in the industry and society.

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Technology, Environment & Society

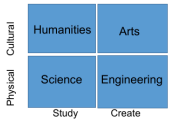
Course Contents

- Definition of terminology – technology, environment, society and development; Inter-dependence of technology, environment, society and development;
- Growth of technologies and its contribution to human development; Current state of technology and its future use as an instrument of change in twenty first century;
- Impact of technology upon the environment, impact of the environment upon human changes in the global climates;
- Environment friendly technology, Technology and development; Renewable energy and environments;
- Technology and environment hazards, its remedy. Major hazards of industry. The improvement of working conditions in the industry.

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Technology, Environment & Society

Engineering and Technical education



Engineering is the discipline focused on **creating our physical world**.

Projection of different disciplines onto a plane defined by 'cultural' vs. 'physical' and 'study' vs. 'create'.

(Src. Kapranos 2012).

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Technology, Environment & Society

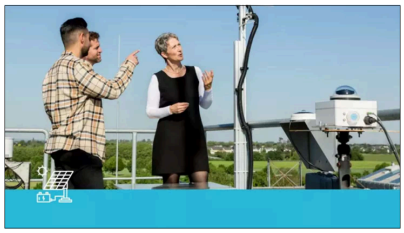
Introduction

'What do we envisage the engineers/technologists of the 21st century ought to be like'?

'What competences (knowledge, skills, attitudes) should they possess'?



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Technology, Environment & Society

Video1 Video1



Q: Watch these Videos and describe how technology and environment are interdependent to each other. Does technology change the ways of living/ lifestyle? Explain How?

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After completing this session, you will be able to explain the interdependency among

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Technology, **Environment**, society and development

From Wikipedia, the free encyclopedia

Environment most often refers to:

Natural environment/Biophysical environment, referring respectively to all living and non-living things occurring naturally and the physical and biological factors along with their chemical interactions that affect an organism or a group of organisms

https://en.wikipedia.org/wiki/Environment

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After completing this session, you will be able to explain the interdependency among

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Technology, Environment, **Society** and development

Society (PEOPLE)

- ✓ **a large group of people who live together in an organized way, making decisions about how to do things and sharing the work that needs to be done. All the people in a country, or in several similar countries, can be referred to as a society**

i.e.We must also consider the needs of the younger/older ***members of society.***

- ✓ **people considered as a group, or a group of people who live together in a particular social system:**

i.e. Societies change over the course of time.

https://dictionary.cambridge.org/dictionary/english/society
https://en.wikipedia.org/wiki/Technology_and_society

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After completing this session, you will be able to explain the interdependency among

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Technology, Environment, **Society** and **Development**

Assignment 1:
Explain how technology, environment, society and development are inter-dependent to each other. You may also refer to the videos showed in slide 13.

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Technology, Environment, **Society** and **Development**

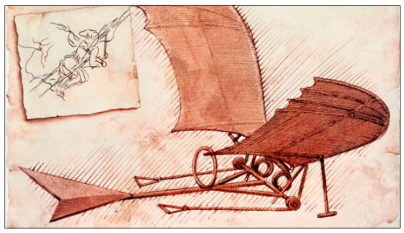
History of Technology Timeline

3.3 million years ago: The first tools
6000 BCE: Irrigation
4000 BCE: Sailing
850 CE: Gunpowder

1 million years ago: Fire
1200 BCE: Iron
950: Windmill
1044: Compass
1250–1300: Mechanical clock
1455: Printing
1765: Steam engine
1804: Railways

https://www.britannica.com/story/history-of-technology-timeline

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Environment and the Earth System

Islamic University of Technology | Hasladar | Friday, October 4, 2024

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Environment and Earth System

Environment

The environment is the aggregate of all external factors or conditions that influence the activities and existence of all living beings including human. Or
The environment is the interaction within and among physical, living and human world in our surrounding.

Surroundings!!!
- A certain distance?
- the room; we live in?
- building?
- City?
- District or division?

-Country?
- Planet?

http://www.yourarticlelibrary.com/essay/environment-and-its-components-explained-with-diagram/44682

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Environment and Earth System

The environment has three important constituents or worlds:

1. Physical world/ Abiotic; consists of nonliving objects (e.g. Rocks, soil, cloud, air) and processes (e.g. Evaporation, season, weather, climate etc.).

This **physical constituent** of the environment is again divided into three parts: (i) **Atmosphere** (gas), (ii) **Hydrosphere** (liquid), and (iii) **Lithosphere** (solid)

2. Living world/ Biotic; consists of living things (e.g. millions of plants, animals, fungus and bacteria's) and life process (e.g. feeding, reproduction species, respiration and photo-synthesis, etc.)

3. Human world/Social; consists of all things that human make, do and believe (e.g. science, arts, religion, literature etc.)

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Environment and the Earth System

Q: Do these three 'worlds' exist in total isolation from one another?

Ans: Of course not!!!

Interaction between PW, LW and HW

Inhale ⟶ Air (PW)
Eat ⟶ Food (LW)

http://www.yourarticlelibrary.com/essay/environment-and-its-components-explained-with-diagram/44682

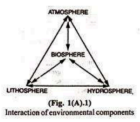
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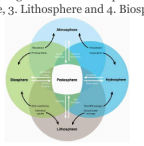
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Environment and the Earth System

The three components of the environment give rise to four important zones: 1. Atmosphere, 2. Hydrosphere, 3. Lithosphere and 4. Biosphere.



(Fig. 1(A).1)
Interaction of environmental components



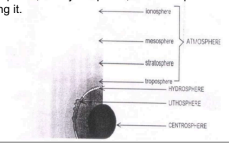
There is continuous interaction among these four zones.

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Environment and Earth System

Earth System

The earth is a complex system consisting of different components; the lithosphere, the hydrosphere, the atmosphere and the biota inhabiting it.



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Ecological Succession

A biome is determined mainly by its climate-like temperature and rainfall.
The earth's land areas are divided into 6 major biomes.

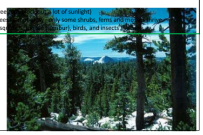
1. TUNDRA

- Bitterly cold and covered with snow and ice.
- Ground remains permanently frozen (permafrost).
- Only certain small plants such as mosses and grasses can grow.



2. CONIFEROUS FOREST (TAIGA)

- Conifers = cone-bearing trees (pines, fir)
- Cold climate
- Dense forests (trees [illegible] lot of sunlight)
- Grasses, small trees [illegible] some shrubs, ferns and [illegible]
- Animals such as [illegible] bear, birds, and insects [illegible]



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Ecological Succession

3. DECIDUOUS (TEMPERATE) FOREST

- Thrive in moderate climates.
- Summers may be hot and winters may be cold.
- Dense forests
- Provides habitats for many animals.



4. TROPICAL RAIN FOREST

- Excellent for plant growth and soil development.
- Plants grow thick and tall.
- Found in areas near the equator.
- Support more plant life and animal species than any other biome.



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Ecological Succession

5. GRASSLANDS

- Mostly grass!
- Temperatures similar to deciduous forest.
- Do not receive much rainfall –enough rain to support grasses but not trees.



6. DESERT

- Very dry; receives very little rainfall.
- Very hot during the day, cold at night.
- Soil is very dry and poor.
- Only a few kinds of plants grow in the desert.
- Very few animals can survive in the desert.



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Ecological Succession

Species

A group of organisms that can breed to produce fully fertile offspring



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Ecological Succession

Population

A group of organism of the same species which live in the same habitat at the same time where they can freely interbreed.



The black-veined white butterfly (*Aporia crataegi*) mating

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Biodiversity

The total number of different species in an ecosystem and their relative abundance





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Structure and Function of Ecosystem

Vertical distribution species occupying different levels is called stratification. For instance, trees canopy is top layer of a forest while grasses are at the bottom.

Functions:

a) Productivity b) Decomposition c) Energy Flow and d) Nutrient Cycle

a) Productivity

Biomass: Mass of living biological organism in a given area at a given time

species biomass-mass of one species

community biomass- mass of all species in the community (including microorganisms, plants or animals). It is measured either in terms of dry weight (60-80% weight of water not included) or by measuring organically bound carbon.

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Structure and Function of Ecosystem

Primary Production: Total amount of biomass or organic matter produced per unit area. Unit in terms of weight (gm^{-2}) or energy ($kcal/m^2$)

Primary productivity: Amount of biomass produced per unit area over a time period by plants during photosynthesis ($gm^{-2}y^{-1}$/$kcalm^{-2}yr^{-1}$)
Secondary Productivity is defined as the rate of formation of new organic matter by consumer

b) Decomposition

Decomposers break down complex organic matter into inorganic substances like CO_2, water and nutrients and the process is called decomposition.

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Decomposition

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Structure and Function of Ecosystem

Energy is transferred from one trophic level to another

- First trophic level (T1) = Producer (Phytoplankton, Grass, Trees)
- Second trophic level (T2) = Primary consumer (Zooplankton, rabbit)
- Third trophic level (T3) = secondary consumer (Birds, fishes)
- Fourth trophic level (T4) = top consumer (Man, Lion)

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Structure and Function of Ecosystem

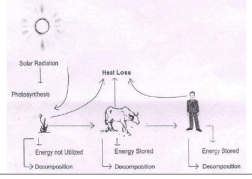
Rule of 10% (10% law of Lindeman): energy is reduced by 10% during flow to every higher trophic level. Because of this law, the number of trophic levels are usually restricted to 4-5.

Right from the sunlight being trapped by autotrophs into organic material, and further transfer of energy to higher trophic levels. On an avegrage, 10% (2-20%) of energy is transferred while rest 90% (80-98%) is lost in respiratory and other losses. So as the number of trophic levels increase in a food chain, most of the energy is lost.

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Energy Flow in the Food Chain



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